MATH70078: Fundamentals of Statistical Inference - Coursework

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Declaration: This is my own and unaid work.

1 Question (a)

(1) The Bayes risk w.r.t prior π is

$$R_1(\pi, d) = \inf_{d \in \mathcal{D}} \int \mathbb{E}(L(\theta, d(Y))\pi(\theta)d\theta, \tag{1}$$

and the Bayes rule is

$$\int L(\theta, d(y))\pi(\theta|y)d\theta. \tag{2}$$

Given each y, it is possible to choose $a \in \mathcal{A}$ to minimize the loss function and obtain the Bayes loss. Hence, we can express the Bayes risk as

$$R_1(\pi, d) = \int \left\{ \underbrace{\inf_{a \in \mathcal{A}} \int L(\theta, a)\pi(\theta|y)d\theta}_{R_0(\pi(\theta|y))} \right\} f(y)dy = \int \left\{ \inf_{a \in \mathcal{A}} \int L(\theta, a) f(\theta, y)d\theta \right\} dy. \tag{3}$$

(2) Given the context, we first explore the objection

$$\int c|S| - \mathbf{I}_S(\theta)d\theta = \int_S c - \pi(\theta)d\theta. \tag{4}$$

To make this objection as small as possible, we need to search the most board interval S^*such that $c \leq \pi(\theta)$, which is denoted as

$$S^* = \{\theta : \pi(\theta) \geq c\}. \tag{5}$$

Hence, the Bayes loss of the Bayes action R_0 is

$$R_0(\pi) = \int_{S^*} c - \pi(\theta)d\theta \leq \int_{S^*} 0d\theta = 0. \tag{6}$$

(3) Suppose the prior of θ is $\mathcal{N}(\mu, \sigma^2)$, then

$$S^* = \{\theta : \mu - \sigma \cdot z < \theta < \mu + \sigma \cdot z\}, \tag{7}$$

where $z(\sigma) = \sqrt{-2\log(c\sqrt{2\pi\sigma^2})}$ and $\sigma > 0$. Hence, we derive that

$$R_0(\sigma) = \int_{s^*} cd\theta - \int_{\mu-\sigma z}^{\mu+\sigma z} \mathcal{N}(\theta; \mu, \sigma^2)d\theta = \int_{s^*} cd\theta - \int_{-z}^{z} \mathcal{N}(\theta; 0, 1)d\theta = 2c\sigma \cdot z - 2\Phi(z) + 1. \tag{8}$$

(4) Let us first figure out R_1. From (2), to optimize this loss function, we need to choose

$$S_y^* = \{\theta : \pi(\theta|y) \geq c\}. \tag{9}$$

According to Bayes's theorem,

$$\pi(\theta|Y) \propto \pi(\theta)f(Y;\theta) \propto \mathcal{N}(\theta; 0, 1)\mathcal{N}(Y; \theta, 1/9) \Rightarrow \pi(\theta|Y) = \mathcal{N}(\frac{9}{10}Y, \frac{1}{10}). \tag{10}$$

But from (8), the Bayes loss of the Bayes action is independent with the mean μ, which is to say the Bayes decision rule is independent with the observation Y but depends on the variance of the prior. The Bayes risk then becomes

$$R_1 = \iint_{S^*} L(\theta, S)\pi(\theta|y)d\theta f(y)dy = \int R_0(\frac{1}{10})f(y)dy = R_0(\frac{1}{10}), \tag{11}$$

but with different σ^2. We need to discuss the monotonicity of R_0. It is a big project to calculate the derivative, however, by differentiate the whole R_0 w.r.t σ, we have

$$\frac{dR_0}{d\sigma} = 2c(\sigma\frac{dz}{d\sigma} + z) - 2\frac{d\Phi}{dz} \cdot \frac{dz}{d\sigma}. \tag{12}$$

All clues point out that we should find the derivative of z w.r.t σ, and fortunately, we derive

$$\frac{dz}{d\sigma} = -\frac{2}{\sigma\sqrt{-2(\log c\sqrt{2\pi\sigma^2})}} = -\frac{1}{\sigma z}. \tag{13}$$

It is also to worth to point out that the derivative of Φ at $z(\sigma)$ is $c\sigma^2$, then we plug all the parameters in (12), we have

$$\frac{dR_0}{d\sigma} = 2cz > 0, \tag{14}$$

which is to say R_0 is increasing w.r.t σ. Hence, $R_1 = R_0(\frac{1}{10}) < R_0(1) = R_0$.

2 Question (b)

(1) Start with $t \cdot \mathbb{P}(|X| > t)$, then we have a series of deduction:

$$t \cdot \mathbb{P}(|X| > t) = t \cdot \mathbb{E}(\mathbf{1}_{\{|X|>t\}}) \leq \mathbb{E}(|X|\mathbf{1}_{\{|X|>t\}}) = 2\mathbb{E}(X\mathbf{1}_{\{X>t\}})$$
$$= 2\int_t^\infty x\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)dx = -\frac{2}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)\Bigg|_t^\infty = 2\phi(t) \Rightarrow \mathbb{P}(|X| > t) \leq 2\phi(t)/t.$$

(2) We refer the inequality provided:

$$\mathbb{P}(\hat{\theta}_H = 0|\theta = 0) = \prod_{i=1}^n \mathbb{P}(|Y_i| \leq \lambda|\theta_i = 0) = \prod_{i=1}^n \{1 - \mathbb{P}(|Y_i| > \lambda|\theta_i = 0)\}$$
$$\geq \prod_{i=1}^n \{1 - \frac{2\phi(\lambda)}{\lambda}\} = \prod_{i=1}^n (1 - \frac{2\frac{1}{\sqrt{2\pi}}}{n\sqrt{2\log n}})$$
$$= (1 - \frac{2\frac{1}{\sqrt{2\pi}}}{n\sqrt{2\log n}})^n \geq 1 - \frac{2\frac{1}{\sqrt{2\pi}}}{n\sqrt{2\log n}} \to 1$$

As $\mathbb{P}(\hat{\theta}_H = 0|\theta = 0) \leq 1$, then we conclude

$$\mathbb{P}(\hat{\theta}_H = 0|\theta = 0) \to 1, \ \ as\ n\ \to \infty. \tag{15}$$

3 Question (c)

In the case that for any θ, the risk could be expressed as:

$$R(\hat{\theta}_s) = \int_{-\infty}^{-\lambda}(y + \lambda - \theta)^2\phi(y;\theta)dy + \int_{-\lambda}^{\lambda}\theta^2\phi(y;\theta)dy + \int_{\lambda}^{\infty}(y - \lambda - \theta)^2\phi(y;\theta)dy \tag{16}$$
$$= \int_{-\infty}^{-\lambda-\theta}(z + \lambda)^2\phi(z)dz + \theta^2\int_{-\lambda-\theta}^{\lambda-\theta}\phi(z)dz + \int_{\lambda-\theta}^{\infty}(z - \lambda)^2\phi(z)dz. \tag{17}$$

Suppose Φ is the cumulative distribution function of standard normal random variables, it is possible to derive the expression of risk in terms of ϕ and Φ. More specificly, we have

$$\int_{-\infty}^{-\lambda-\theta}(z + \lambda)\phi(z)dz = \mathbb{E}(Z^2\mathbf{1}_{\{Z<-\lambda-\theta\}}) + 2\lambda\mathbb{E}(Z\mathbf{1}_{\{Z<-\lambda-\theta\}}) + \lambda^2\mathbb{E}(\mathbf{1}_{\{Z<-\lambda-\theta\}}) \tag{18}$$
$$= (\lambda + \theta)\phi(-\lambda - \theta) + \Phi(-\lambda - \theta) - 2\lambda\phi(-\lambda - \theta) + \lambda^2\Phi(-\lambda - \theta) \tag{19}$$
$$= (\theta - \lambda)\phi(-\lambda - \theta) + (\lambda^2 + 1)\Phi(-\lambda - \theta), \tag{20}$$

and similarly, we have

$$\int_{\lambda-\theta}^{\infty}(z - \lambda)\phi(z)dz = \mathbb{E}(Z^2\mathbf{1}_{\{Z>\lambda-\theta\}}) - 2\lambda\mathbb{E}(Z\mathbf{1}_{\{Z>\lambda-\theta\}}) + \lambda^2\mathbb{E}(\mathbf{1}_{\{Z>\lambda-\theta\}}) \tag{21}$$
$$= -(\theta + \lambda)\phi(\lambda - \theta) + (\lambda^2 + 1)(1 - \Phi(\lambda - \theta)). \tag{22}$$

Hence, the risk could be expressed in terms of ϕ and Φ, which is

$$R(\hat{\theta_s}) = (\theta-\lambda)\phi(-\lambda-\theta)+(\lambda^2+1)\Phi(-\lambda-\theta)-(\theta+\lambda)\phi(\lambda-\theta)+(\lambda^2+1)(1-\Phi(\lambda-\theta))+\theta^2(\Phi(\lambda-\theta)-\Phi(-\lambda-\theta)). \tag{23}$$

We will use the similar idea and the Stein's Lemma to show the upper bound when $\theta = 0$. However, it is important to note that

$$g(Y) = \frac{(Y-\lambda)^2\mathbf{1}_{\{Y\geq\lambda\}}}{Y} \tag{24}$$

is continuously differentiable, as $g'(Y) = (1-\frac{\lambda^2}{Y^2})\mathbf{1}_{\{Y\geq\lambda\}}$ is continuous on the $(-\infty,\infty)$, which satisfies the setting for Stein's Lemma. Therefore, we have

$$\begin{aligned} R(\hat{\theta}_s) = \mathbb{E}(\hat{\theta}_s^2) &= \mathbb{E}\left[(Y-\lambda)^2\mathbf{1}_{\{Y\geq\lambda\}}\right] + \mathbb{E}\left[(Y+\lambda)^2\mathbf{1}_{\{Y\leq-\lambda\}}\right] = 2\mathbb{E}\left[(Y-\lambda)^2\mathbf{1}_{\{Y\geq\lambda\}}\right] \\ &= 2\mathbb{E}\left[\frac{(Y-\lambda)^2}{Y}Y\mathbf{1}_{\{Y\geq\lambda\}}\right] = 2\mathbb{E}\left[(1-\frac{\lambda^2}{Y^2})\mathbf{1}_{\{Y\geq\lambda\}}\right] = \frac{2}{\lambda}\mathbb{E}\left[(\lambda-\frac{\lambda^3}{Y^2})\mathbf{1}_{\{Y\geq\lambda\}}\right] \\ &\leq \frac{2}{\lambda}\mathbb{E}\left[\lambda\mathbf{1}_{\{Y\geq\lambda\}}\right] \leq \frac{2}{\lambda}\mathbb{E}\left[Y\mathbf{1}_{\{Y\geq\lambda\}}\right] = \frac{2\phi(\lambda)}{\lambda}. \end{aligned}$$

The alternative solution is motivated by using the expression of ϕ and Φ. When $\theta = 0$, we could simply the risk as

$$R(\hat{\theta_s}) = (1+\lambda^2)\Phi(-\lambda) - \lambda\phi(-\lambda) - \lambda\phi(\lambda) + (1+\lambda^2)(1-\Phi(\lambda)) \tag{25}$$

$$\leq (1+\lambda^2)\mathbb{P}(|Z|>\lambda) - 2\lambda\phi(\lambda) \leq (2+2\lambda^2)\phi(\lambda)/\lambda - 2\lambda^2\phi(\lambda)/\lambda = 2\phi(\lambda)/\lambda. \tag{26}$$

As $\theta \to +\infty$, the first term in (17) goes to 0 as we can regard

$$\int_{-\infty}^{-\lambda-\theta}(z+\lambda)^2\phi(z)dz = \int_{-\infty}^{-\infty}(z+\lambda)^2\phi(z)dz = 0;$$

however the third term, we have

$$\int_{\lambda-\theta}^{\infty}(z-\lambda)^2\phi(z)dz = \int_{-\infty}^{\infty}(z-\lambda)^2\phi(z)dz = \mathbb{E}(Z^2) + 2\cdot\lambda\mathbb{E}(Z) + \lambda^2 = 1+\lambda^2.$$

The most tricky part is the second term, define the quant

$$f(\theta) = \frac{\int_{-\lambda-\theta}^{\lambda-\theta}\phi(z)dz}{1/\theta^2}, \tag{27}$$

which is $\frac{0}{0}$ as $\theta \to +\infty$. We attempt to use L'Hospital rule:

$$\lim_{\theta\to\infty} f(\theta) = \lim_{\theta\to\infty} \frac{\phi(\lambda+\theta) - \phi(\lambda-\theta)}{-2/\theta^3} = 0 \tag{28}$$

because of the fact the the exponential function is decreasing faster than the rational functions. As λ becomes negligible compared to extremely large θ, the lower limit of integral goes to $-\infty$. Hence, it is expected that the Bayes risk $R(\hat{\theta}_s)$ will converge to $1 + \lambda^2$. By symmetry, when $\theta \to -\infty$, the Bayes risk will converge to $1 + \lambda^2$ either.

Let $\hat{Y}$ is the unbiased estimator for θ, then the risk is determined:

$$\mathbb{E}(\hat{Y} - \theta)^2 = Var(\hat{Y}) = 1. \tag{29}$$

We test the cases when $\lambda = 1, 2, 5$:

- $\lambda = 1$, then the threshold $|\theta| \leq 1.283264$ will achieve the risk less than 1,
- $\lambda = 2$, then the threshold $|\theta| \leq 1.052787$ will achieve the risk less than 1,
- $\lambda = 5$, then the threshold $|\theta| \leq 1.000006$ will achieve the risk less than 1.

The soft threshold estimator is preferred when its risk is less than that of the unbiased estimator, whose preferable interval of θ is given above.